You **cannot directly know intention**, but you **can detect patterns that strongly indicate intentional wrong answers**. The trick is to use **behavior analysis + consistency checks**.

---

1. Repeat Question Consistency Test

If a child answers randomly or intentionally wrong, their answers become **inconsistent**.

Implementation

Show the **same letter again later** without telling them.

Example sequence:

Round	Letter	Child Answer
1	E	P
4	E	E

This means:

- child **can see the letter**
- first answer was **not genuine**

Reliability Logic

```
if same_letter_shown_again:
   if previous_answer != new_answer:
      reliability_score -= 10
```

If **many inconsistencies occur → intentional behaviour likely**.

---

2. Response Time Analysis

Children who actually read letters take **~1–2 seconds**.

Intentional/random answering often gives:

- **very fast responses (<0.4 sec)**
- repeated identical timing

Implementation

```
response_time = answer_time - letter_display_time
```

Detection rule:

Response Time	Meaning
<0.4 sec	guessing

Response Time	Meaning
0.8–3 sec	normal reading
>5 sec	struggling / confusion

Flag suspicious pattern:

```
if response_time < 0.4:
   guessing_count += 1
```

---

3. Difficulty Pattern Detection

Human vision follows a **logical pattern**.

Correct pattern:

Large letters → correct

Medium letters → few mistakes

Small letters → many mistakes

Intentional wrong answers produce **impossible patterns**.

Example:

Size	Result
Large	Wrong
Medium	Correct
Small	Correct

This is **biologically unlikely**.

Detection:

```
if large_letter_wrong and small_letter_correct:
   reliability_score -= 15
```

---

4. Eye Tracking Validation

Use gaze tracking to confirm the child **actually looks at the letter**.

Using:

- MediaPipe
- OpenCV

Check:

- gaze direction
- fixation time

Logic:

```
if gaze_not_on_letter:
    answer_valid = False
```

If the child **never looked at the letter but answered**, it's likely guessing.

---

5. Random Guess Probability Detection

Your letters set may contain:

[C D E F L O P T Z]

Total options = **9**

Random guessing accuracy ≈ **11%**

If a child gets:

60% correct

but inconsistent pattern

This means **answers are intentional or assisted**.

---

6. Attention Monitoring

Track:

- head movement
- gaze stability
- blink pattern

Using:

- MediaPipe Face Mesh

Example rule:

```
if gaze_off_screen > 40%:
    test_reliability = low
```

---

7. Hidden Verification Questions

Insert **easy control letters** randomly.

Example:

Large letter: O

If child says **wrong**:

child likely not cooperating

Logic:

```
if very_large_letter_wrong:
    reliability_score -= 20
```

---

8. Reliability Score System

Create a **test trust score**.

Example:

score = 100

inconsistency → -10

fast guess → -5

gaze off screen → -10

impossible pattern → -15

control letter wrong → -20

Final:

Score	Interpretation
80–100	Reliable
60–80	Moderate reliability
<60	Unreliable test

---

9. Final Decision Logic

```
if reliability_score < 60:
    show_message("Test unreliable. Please repeat test.")
```

This prevents **false medical conclusions**.

---

10. Important Reality Check

Even hospitals cannot **100% detect intention**.

They rely on:

- **behavior pattern analysis**
- **consistency**
- **attention tracking**

Your system should do the same.

Most student projects stop at **"check correct or wrong answers."** That's weak. Anyone can randomly click and your system becomes useless.

The **advanced trick** is to detect **behavior patterns instead of just answers**.

The idea:
Build a **behavior-based reliability model** that predicts whether the child is **cooperating, guessing, or intentionally giving wrong answers**.

This makes your system look **10× more advanced in interviews**.

---

1. Build a Behaviour Dataset

Instead of training ML on **letters**, train it on **test behaviour**.

Each test session becomes one row.

Example features:

Feature	Meaning
accuracy	% correct answers
avg_response_time	average time to answer
response_variance	variation in response time
repeat_consistency	repeated letter consistency
gaze_focus_score	how long child looked at letter
blink_rate	eye fatigue indicator
distance_variation	movement during test
large_letter_accuracy	accuracy on easy letters
small_letter_accuracy	accuracy on hard letters

Example row:

accuracy = 0.55

avg_response_time = 0.3

repeat_consistency = 0.4

gaze_focus_score = 0.35

large_letter_accuracy = 0.3

small_letter_accuracy = 0.7

This pattern is **suspicious**.

---

2. Define Behaviour Classes

Train a classifier with labels like:

Label	Meaning
Cooperative	child trying honestly
Guessing	random answers
Not Paying Attention	looking away
Intentional Wrong	purposely incorrect

---

3. Train a Behaviour Classifier

Use:

- Scikit-learn
- Random Forest (best for tabular data)

Example:

```
from sklearn.ensemble import RandomForestClassifier

model = RandomForestClassifier()

model.fit(X_train, y_train)
```

Prediction:

Behavior = Guessing

Confidence = 82%

---

4. Add Control Letters (Psychological Trick)

Real eye tests do this.

Insert **obvious letters** randomly.

Example:

Sequence

Row1: E

Row2: F

Row3: O

Row4: P

Row5: HUGE LETTER Z

If the child misses the **huge letter**, something is wrong.

---

5. Response Time Curve Analysis

Reading letters follows a **curve**.

Example normal pattern:

Large letters → fast response

Small letters → slower response

Intentional answers look like:

Large letters → wrong + fast

Small letters → correct

Impossible pattern.

---

6. Gaze-Answer Synchronization (VERY ADVANCED)

Track **eye fixation before answering**.

Using:

- MediaPipe
- OpenCV

Logic:

child must look at letter

for at least 300–500 ms

before answering

If answer comes **without gaze fixation**, it's guessing.

---

7. Entropy of Answers (Statistics Trick)

If answers look **too random**, detect it.

Example letter set:

C D E F L O P T Z

Random guessing produces **high entropy distribution**.

Example:

C: 2

D: 3

E: 1

F: 2

L: 2

Very uniform → guessing.

---

8. Reliability Score (Final Layer)

Combine everything.

Example formula:

Reliability =

0.3 * consistency +

0.2 * gaze_focus +

0.2 * response_pattern +

0.2 * control_letter_accuracy +

0.1 * distance_stability

Output:

Reliability = 0.78

Test Valid

or

Reliability = 0.42

Test Invalid

---

9. Industry-Level Trick

Add **Bayesian Confidence Score**.

Instead of saying:

Vision risk = Moderate

Say:

Vision risk = Moderate

Confidence = 83%

Test reliability = 76%

That makes your system look **clinical-grade**.

---

10. Why This Impresses Interviewers

Most student projects do:

show letter

check answer

calculate score

Your project would do:

computer vision

eye tracking

behavior analysis

machine learning

test reliability modeling

That's **AI + CV + ML + psychology**.

Very few students think that far.